Exp 3: Benchmarking of 5G/6G Communication for IoE Applications based on Throughput, Latency, Reliability and Energy Efficiency

Objective:

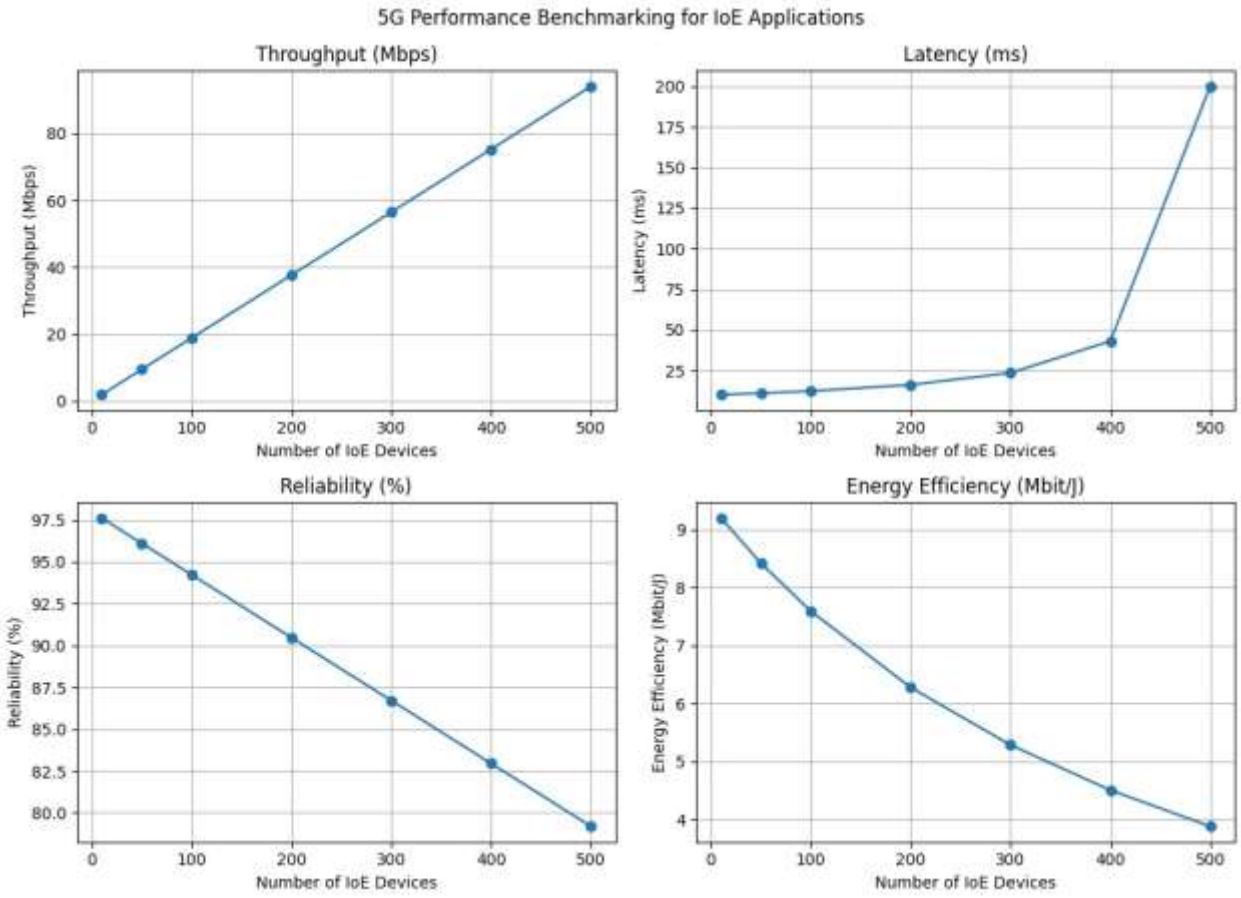
To benchmark and compare 5G and 6G communication technologies for Internet of Everything (IoE) applications based on throughput, latency, reliability, and energy efficiency

Procedure :

1. Identify suitable IoE applications and select 5G and 6G communication technologies for comparison.
2. Select the performance parameters: throughput, latency, reliability, and energy efficiency.
3. Collect or simulate 5G and projected 6G performance data.
4. Compare both technologies using tables and graphs based on the selected parameters.
5. Analyze the results to determine the suitability of 5G and 6G for different IoE applications.

Tabular column:

IoE Devices	220	350	480
Offered Load	42.24 Mbps	67.20 Mbps	92.16 Mbps
Throughput	41.40 Mbps	65.86 Mbps	90.32 Mbps
Latency	17.31 ms	30.49 ms	127.55 ms
Reliability	89.72 %	84.83 %	79.94 %
Energy Efficiency	6.06 Mbit/J	4.87 Mbit/J	3.99 Mbit/J



Result :

As the number of IoE devices increases, throughput increases, while latency, reliability, and energy efficiency decrease. Therefore, 5G is suitable for moderate-load IoE applications, whereas 6G offers higher throughput for high-demand applications but requires improvements in latency, reliability, and energy efficiency.